

CureAgent: A Training-Free Executor-Analyst Framework for Clinical Reasoning

NEURAL INFORMATION
PROCESSING SYSTEMS

Ting-Ting Xie and Yixin Zhang
Team: Cure Agent

Introduction

The Problem: Context Utilization Failure

- Current clinical agents suffer from a critical **"Context Utilization Failure"**: models successfully retrieve evidence but fail to ground their diagnosis in that information.
- Monolithic models (e.g., Llama-3-8B) struggle to balance the syntactic precision required for tool execution with the semantic reasoning robustness needed for clinical logic.

The Gap

- Open-Source Gap**: Standard open-source models (Llama-3.1, Qwen3) achieve zero-shot scores below 30% on phase1, lacking specific syntax training.

model name	phase1 [†]	phase2
TxAgent [8]	65.214	69.325
Llama-3.1-8B [9]	27.912	29.715
Llama3-Med42-8B [10]	46.125	50.407
Qwen3-8B [11]	28.134	36.264
Qwen-3-32B-Medical-Reasoning [12]	36.513	40.791
Baichuan-M2 [13]	29.944	30.071
medgemma-4b-it [14]	50.331	54.366
gpt-oss-20b [15]	45.425	47.644
gpt-oss-120b [15]	45.377	39.896

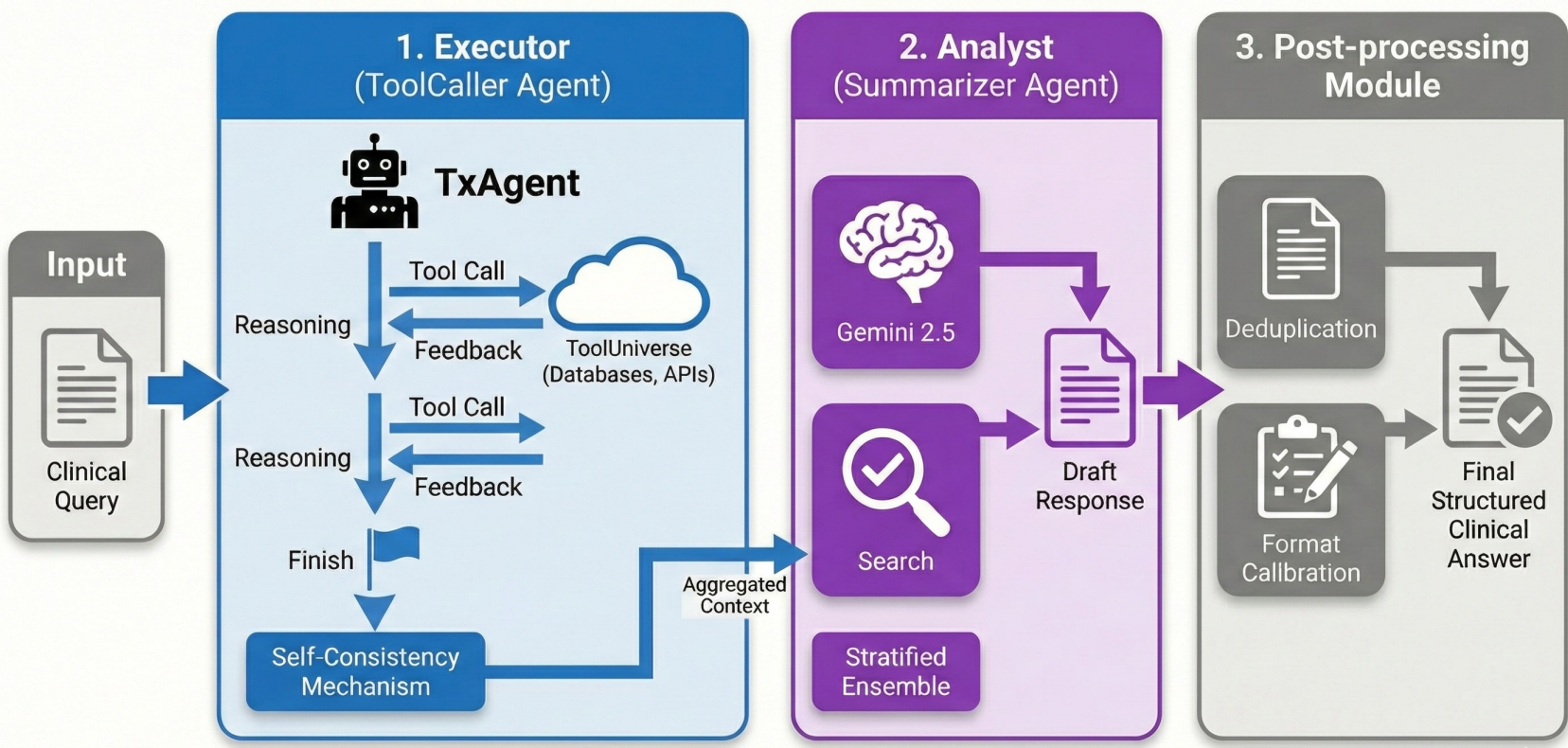
- Closed-Source Ceiling**: Even powerful reasoners like Gemini 2.5 Pro hit a performance ceiling (~75%) without specialized tool handling.

model name	with tools	phase2
gemini-2.5-flash	×	63.104
gemini-2.5-flash	search	69.627
gemini-2.5-pro	×	67.753
gemini-2.5-pro	search	74.806
gemini-3-pro-preview	×	70.985
gemini-3-pro-preview	search	81.283

Our Solution: CureAgent

- Executor-Analyst Framework**, decoupling "The Hand" (Tool Execution) from "The Brain" (Clinical Reasoning).
- Achieves **2nd place (83.8%)** on CURE-Bench without expensive end-to-end retraining.
- Training-Free Agility**: Our design allows for "hot-swapping" the reasoning backbone (e.g., upgrading to Gemini 3 Pro) without retraining the specialized tool-use model.

Methods



The Executor ("The Hand")

- TxAgent (Llama-3.1-8B fine-tuned) navigates the "ToolUniverse" (200+ tools). It utilizes Self-Consistency (n=10) to aggregate top-k evidence and reasoning trace, ensuring robust retrieval.

The Analyst ("The Brain")

- Gemini 2.5 performs pure clinical reasoning. Freed from syntactic tool burdens, it focuses on context synthesis, information supplementation and fact verification with search.

Topology: Stratified Ensemble

- We partition agents into isolated pipelines (each with 10 Executors to 1 Analyst). By aggregating the Analysts' final outputs via majority vote, this topology preserves evidentiary diversity and prevents premature information filtering.

Results

State-of-the-Art Performance

- CureAgent achieves 81.37% accuracy using gemini-2.5-flash without search (see Table below), and 83.8% with search on leaderboard, significantly outperforming the competition baseline TxAgent.
- Decoupling Effect**: Combining a single Executor with a single Analyst jumps performance from 69.3% (TxAgent alone) to 74.7%.

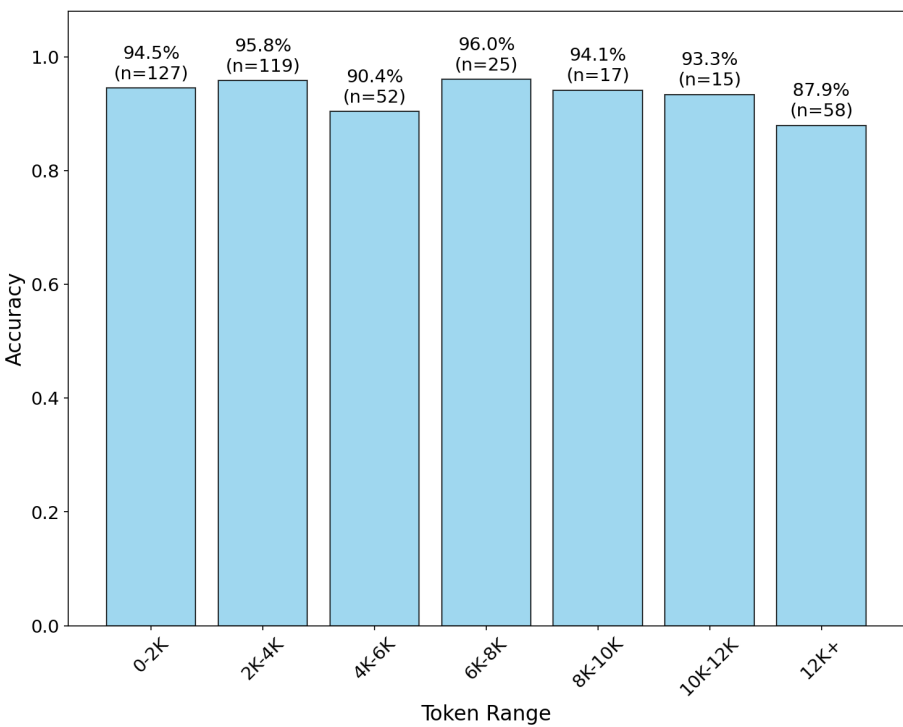
Agent1	n ₁	Agent2	n ₂	phase2 (%)
<i>Baselines</i>				
gemini-2.5-flash	1	-	-	63.104
TxAgent	1	-	-	69.325
<i>Single Agent with Self-Consistency</i>				
gemini-2.5-flash	3	-	-	65.678
TxAgent	10	-	-	72.673
TxAgent	30	-	-	73.508
<i>Collaborative</i>				
TxAgent	1	gemini-2.5-flash	1	74.701
TxAgent	10	gemini-2.5-flash	1	78.527
TxAgent	30	gemini-2.5-flash	1	79.311
TxAgent	30	gemini-2.5-flash	3	80.510
TxAgent	10*3	gemini-2.5-flash	3	81.367
TxAgent	10*3	gemini-2.5-flash (with search)	3	83.803

Ablation: Topology Matters

- Stratified Ensemble (Late Fusion)** outperforms Global Pooling (Early Fusion) by ~1% (81.51%→81.37%), proving that preserving evidentiary diversity matters.

(Open-Challenge) The Context-Performance Paradox

- Extending reasoning contexts beyond 12k tokens introduces noise that degrades accuracy (dropping from **94% to 87.9%**).



(Open-Challenge) Curse of Dimensionality in Action Spaces

- Insight: Expanding the tool library from 200+ to 600+ tools caused a performance drop from **92.0% to 87.5%** on validation set.

model	ToolUniverse 1.0 (200+ tools)	ToolUniverse 2.0 (600+ tools)
TxAgent	92.0%	87.5%

Conclusions

- Decoupling Solves "Context Failure"**: We validated that separating syntactic tool precision ("The Hand") from semantic clinical reasoning ("The Brain") effectively bridges the gap between retrieval and diagnosis.
- Topology Matters**: Our ablation studies prove that high-fidelity decision-making requires **preserving evidentiary diversity**. The Stratified Ensemble (Late Fusion) is essential to prevent early information bottlenecks.
- Architectural Agility**: We achieved SOTA performance via **training-free architectural engineering**. This proves that optimizing interaction design (e.g., hierarchical orchestration, early rejection) is a more scalable alternative to expensive end-to-end retraining for evolving clinical AI.
- Addressing Scaling Limits**: To tackle the "Curse of Dimensionality" in massive toolsets and the "Context-Performance Paradox" (accuracy drops >12k tokens), future architectures must pivot to **Hierarchical Tool Indexing** and confidence-based **Early Rejection** to filter noise before reasoning.

Resources & Acknowledgments

More details could be found in **Tech report & Code**:

<https://github.com/June01/CureAgent>

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Email: helloxietingting@gmail.com & zyxcambridge@gmail.com